ACTIVIDAD 8

DNS

Primeramente entraremos en el archivo named.conf.local para configurarlo

```
dani@dani-Standard-PC-i440FX-PIIX-1996: ~
dani@dani-Standard-PC-i440FX-PIIX-1996:~$ sudo nano /etc/bind/named.conf.local
```

Dentro del archivo pondremos las siguientes lineas:

```
  GNU nano 2.5.3        Archivo: /etc/bind/named.conf.local          Modificado

//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "iesmarisma.intranet" {
        type master;
        file "/etc/bind/zones/db.iesmarisma.intranet";
};

zone "informatica.iesmarisma.intranet" {
        type master;
        file "/etc/bind/zones/db.informatica.iesmarisma.intranet";
};
```

Ahora crearemos el directorio para zonas y tambien editaremos la zona principal

```
dani@dani-Standard-PC-i440FX-PIIX-1996: ~
dani@dani-Standard-PC-i440FX-PIIX-1996:~$ sudo mkdir -p /etc/bind/zones
dani@dani-Standard-PC-i440FX-PIIX-1996:~$ sudo nano /etc/bind/zones/db.iesmarism
a.intranet
```

Dentro de la zona principal, pondremos lo siguiente:

```
GNU nano 2.5.3   Archivo: /etc/bind/zones/db.iesmarisma.intranet   Modi

$TTL 604800
@   IN  SOA ns1.iesmarisma.intranet. admin.iesmarisma.intranet. (
        2
        604800
        86400
        2419200
        604800 )

@       IN  NS  ns1.iesmarisma.intranet.

ns1     IN  A   192.168.1.10

www     IN  A   192.168.1.20
ftp     IN  A   192.168.1.21
smtp    IN  A   192.168.1.22

; Delegación del subdominio
informatica IN NS ns1.informatica.iesmarisma.intranet.
```

Ahora iremos a editar la zona del subdominio

```
dani@dani-Standard-PC-i440FX-PIIX-1996: ~
dani@dani-Standard-PC-i440FX-PIIX-1996:~$ sudo nano /etc/bind/zones/db.informati
ca.iesmarisma.intranet
```

El contenido de este archivo será el siguiente:

```
GNU nano 2.5.3 Archivo: ...nes/db.informatica.iesmarisma.intranet Modificado

$TTL 604800
@   IN  SOA ns1.informatica.iesmarisma.intranet. admin.informatica.iesmarisma.i$
        2
        604800
        86400
        2419200
        604800 )

@       IN  NS  ns1.informatica.iesmarisma.intranet.

ns1     IN  A   192.168.1.30

www     IN  A   192.168.1.31
ftp     IN  A   192.168.1.32
smtp    IN  A   192.168.1.33
```

Luego haremos todas las comprobaciones de la configuración de todo lo que hemos cambiado.

```
dani@dani-Standard-PC-i440FX-PIIX-1996:~$ sudo named-checkconf
dani@dani-Standard-PC-i440FX-PIIX-1996:~$ sudo named-checkzone iesmarisma.intran
et /etc/bind/zones/db.iesmarisma.intranet
zone iesmarisma.intranet/IN: informatica.iesmarisma.intranet/NS 'ns1.informatica
.iesmarisma.intranet' (out of zone) has no addresses records (A or AAAA)
zone iesmarisma.intranet/IN: loaded serial 2
OK
dani@dani-Standard-PC-i440FX-PIIX-1996:~$ sudo named-checkzone informatica.iesma
risma.intranet /etc/bind/zones/db.informatica.iesmarisma.intranet
zone informatica.iesmarisma.intranet/IN: loaded serial 2
OK
dani@dani-Standard-PC-i440FX-PIIX-1996:~$
```

Luego reiniciamos para aplicar los cambios

```
dani@dani-Standard-PC-i440FX-PIIX-1996: ~
dani@dani-Standard-PC-i440FX-PIIX-1996:~$ sudo systemctl restart bind9
dani@dani-Standard-PC-i440FX-PIIX-1996:~$
```

Luego configuraremos el cliente en “resolv.conf”

```
dani@dani-Standard-PC-i440FX-PIIX-1996: ~
dani@dani-Standard-PC-i440FX-PIIX-1996:~$ sudo nano /etc/resolv.conf
```

Dentro del archivo pondremos esto:

```
GNU nano 2.5.3          Archivo: /etc/resolv.conf

# Dynamic resolv.conf(5) file for glibc resolver(3) generated by resolvconf(8)
#     DO NOT EDIT THIS FILE BY HAND -- YOUR CHANGES WILL BE OVERWRITTEN
nameserver 127.0.1.1
search home.arpa

nameserver 192.168.1.10
```

Luego haremos las comprobaciones del dominio principal:

```
dani@dani-Standard-PC-i440FX-PIIX-1996:~$ dig www.iesmarisma.intranet

; <<>> DiG 9.10.3-P4-Ubuntu <<>> www.iesmarisma.intranet
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 11267
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;www.iesmarisma.intranet.       IN      A

;; AUTHORITY SECTION:
.                       3360    IN      SOA     a.root-servers.net. nstld.verisi
gn-grs.com. 2026040601 1800 900 604800 86400

;; Query time: 2 msec
;; SERVER: 127.0.1.1#53(127.0.1.1)
;; WHEN: Mon Apr 06 20:50:37 CEST 2026
;; MSG SIZE  rcvd: 127

dani@dani-Standard-PC-i440FX-PIIX-1996:~$
```

```
dani@dani-Standard-PC-i440FX-PIIX-1996:~$ dig ftp.iesmarisma.intranet

; <<>> DiG 9.10.3-P4-Ubuntu <<>> ftp.iesmarisma.intranet
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 27114
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;ftp.iesmarisma.intranet.       IN      A

;; AUTHORITY SECTION:
.                       3324    IN      SOA     a.root-servers.net. nstld.verisi
gn-grs.com. 2026040601 1800 900 604800 86400

;; Query time: 2 msec
;; SERVER: 127.0.1.1#53(127.0.1.1)
;; WHEN: Mon Apr 06 20:51:13 CEST 2026
;; MSG SIZE  rcvd: 127

dani@dani-Standard-PC-i440FX-PIIX-1996:~$
```

```
dani@dani-Standard-PC-i440FX-PIIX-1996: ~
dani@dani-Standard-PC-i440FX-PIIX-1996:~$ dig smtp.iesmarisma.intranet

; <<>> DiG 9.10.3-P4-Ubuntu <<>> smtp.iesmarisma.intranet
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 50578
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;smtp.iesmarisma.intranet.      IN      A

;; AUTHORITY SECTION:
.                       3249    IN      SOA     a.root-servers.net. nstld.verisi
gn-grs.com. 2026040601 1800 900 604800 86400

;; Query time: 1 msec
;; SERVER: 127.0.1.1#53(127.0.1.1)
;; WHEN: Mon Apr 06 20:52:28 CEST 2026
;; MSG SIZE  rcvd: 128

dani@dani-Standard-PC-i440FX-PIIX-1996:~$
```

Ademas es importante comprobar el subdominio:

```
dani@dani-Standard-PC-i440FX-PIIX-1996: ~
dani@dani-Standard-PC-i440FX-PIIX-1996:~$ dig www.informatica.iesmarisma.intrane
t

; <<>> DiG 9.10.3-P4-Ubuntu <<>> www.informatica.iesmarisma.intranet
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 18717
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;www.informatica.iesmarisma.intranet. IN        A

;; AUTHORITY SECTION:
.                       3175    IN      SOA     a.root-servers.net. nstld.verisi
gn-grs.com. 2026040601 1800 900 604800 86400

;; Query time: 2 msec
;; SERVER: 127.0.1.1#53(127.0.1.1)
;; WHEN: Mon Apr 06 20:53:42 CEST 2026
;; MSG SIZE  rcvd: 139

dani@dani-Standard-PC-i440FX-PIIX-1996:~$
```

```
dani@dani-Standard-PC-i440FX-PIIX-1996:~$ dig ftp.informatica.iesmarisma.intrane
t

; <<>> DiG 9.10.3-P4-Ubuntu <<>> ftp.informatica.iesmarisma.intranet
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 27597
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;ftp.informatica.iesmarisma.intranet. IN        A

;; AUTHORITY SECTION:
.                       3142    IN      SOA     a.root-servers.net. nstld.verisi
gn-grs.com. 2026040601 1800 900 604800 86400

;; Query time: 1 msec
;; SERVER: 127.0.1.1#53(127.0.1.1)
;; WHEN: Mon Apr 06 20:54:15 CEST 2026
;; MSG SIZE  rcvd: 139

dani@dani-Standard-PC-i440FX-PIIX-1996:~$
```

```
dani@dani-Standard-PC-i440FX-PIIX-1996:~$ dig smtp.informatica.iesmarisma.intran
et

; <<>> DiG 9.10.3-P4-Ubuntu <<>> smtp.informatica.iesmarisma.intranet
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 15302
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;smtp.informatica.iesmarisma.intranet. IN A

;; AUTHORITY SECTION:
.                       3099    IN      SOA     a.root-servers.net. nstld.verisi
gn-grs.com. 2026040601 1800 900 604800 86400

;; Query time: 2 msec
;; SERVER: 127.0.1.1#53(127.0.1.1)
;; WHEN: Mon Apr 06 20:54:58 CEST 2026
;; MSG SIZE  rcvd: 140

dani@dani-Standard-PC-i440FX-PIIX-1996:~$
```

Por ultimo haremos la delegación, usaremos “NS” para obtener el resultado.

```
dani@dani-Standard-PC-i440FX-PIIX-1996:~$ dig informatica.iesmarisma.intranet NS

; <<>> DiG 9.10.3-P4-Ubuntu <<>> informatica.iesmarisma.intranet NS
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 55567
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;informatica.iesmarisma.intranet. IN    NS

;; AUTHORITY SECTION:
.                       3066    IN      SOA     a.root-servers.net. nstld.verisi
gn-grs.com. 2026040601 1800 900 604800 86400

;; Query time: 1 msec
;; SERVER: 127.0.1.1#53(127.0.1.1)
;; WHEN: Mon Apr 06 20:55:31 CEST 2026
;; MSG SIZE  rcvd: 135

dani@dani-Standard-PC-i440FX-PIIX-1996:~$
```